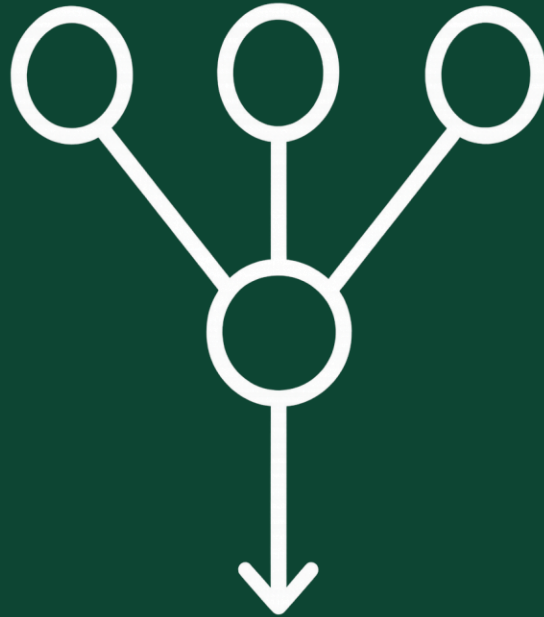


# Market Arbitrage

Queen Mary Machine Learning Society  
Research Trading Division



# What is Market Arbitrage?

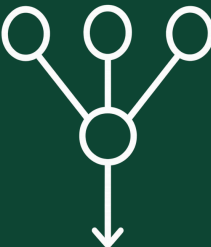
"The practice of **simultaneously buying and selling the same or equivalent asset** in different markets to profit from a **price difference**."

## Key Points:

- Exploits temporary price inefficiencies
- Risk-free in theory when structured correctly
- Requires fast execution and matching trades

## Example Concept:

**Buy an asset for \$0.21** in Market A and sell its equivalent for **\$0.23** in Market B.



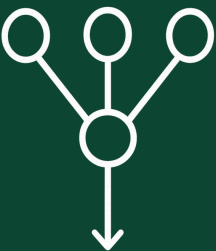
# Why Does it Exist?

## Causes of Price Differences:

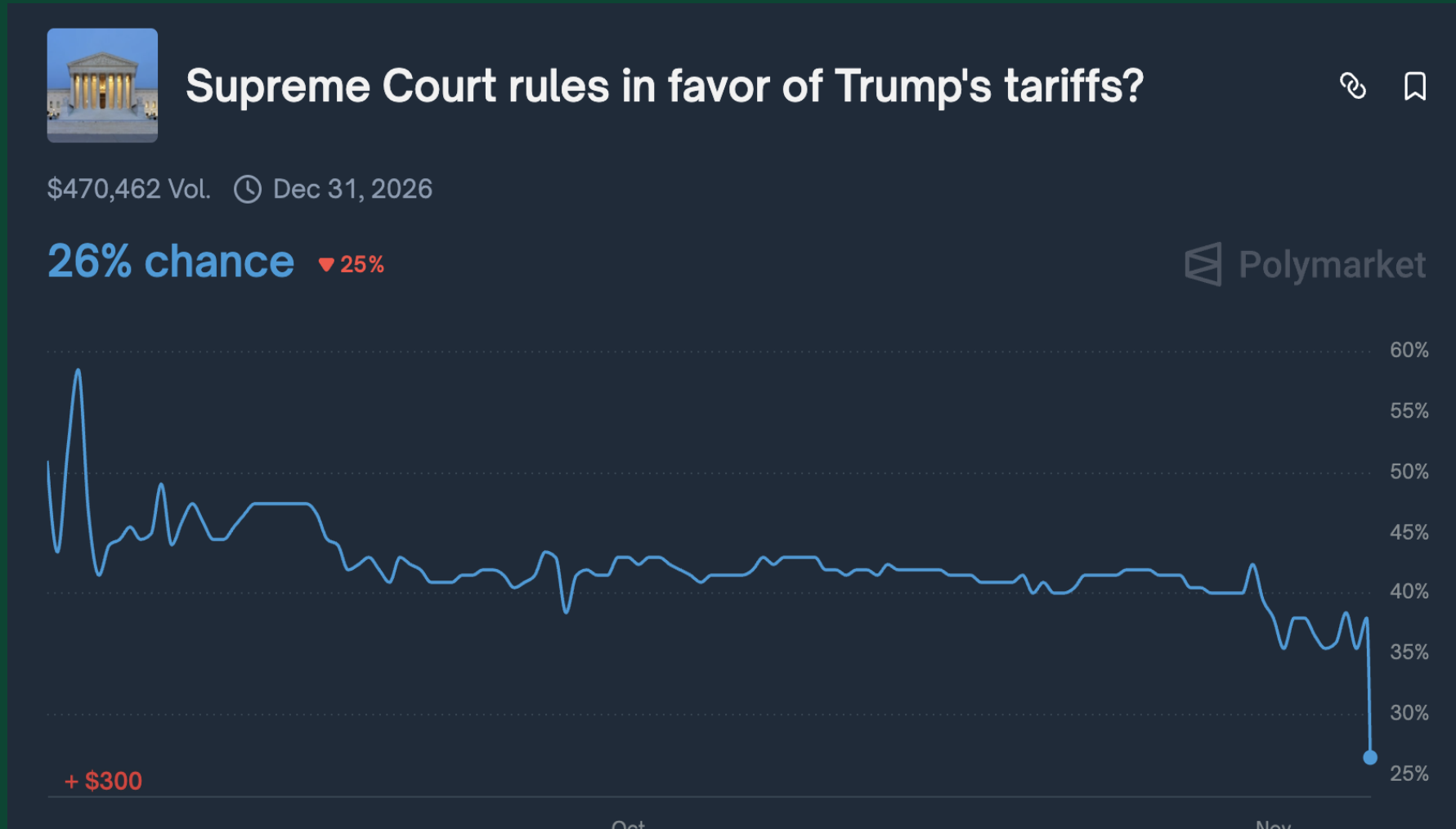
- Market fragmentation (multiple exchanges)
- Information speed differences
- Liquidity imbalances
- Timezone and currency mismatches
- Investor sentiment variations

## Important Insight:

**Arbitrage is a market-correcting force** — traders doing it push prices back into alignment.



# An Example



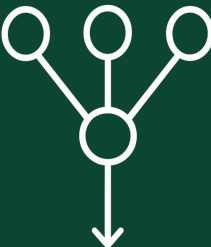
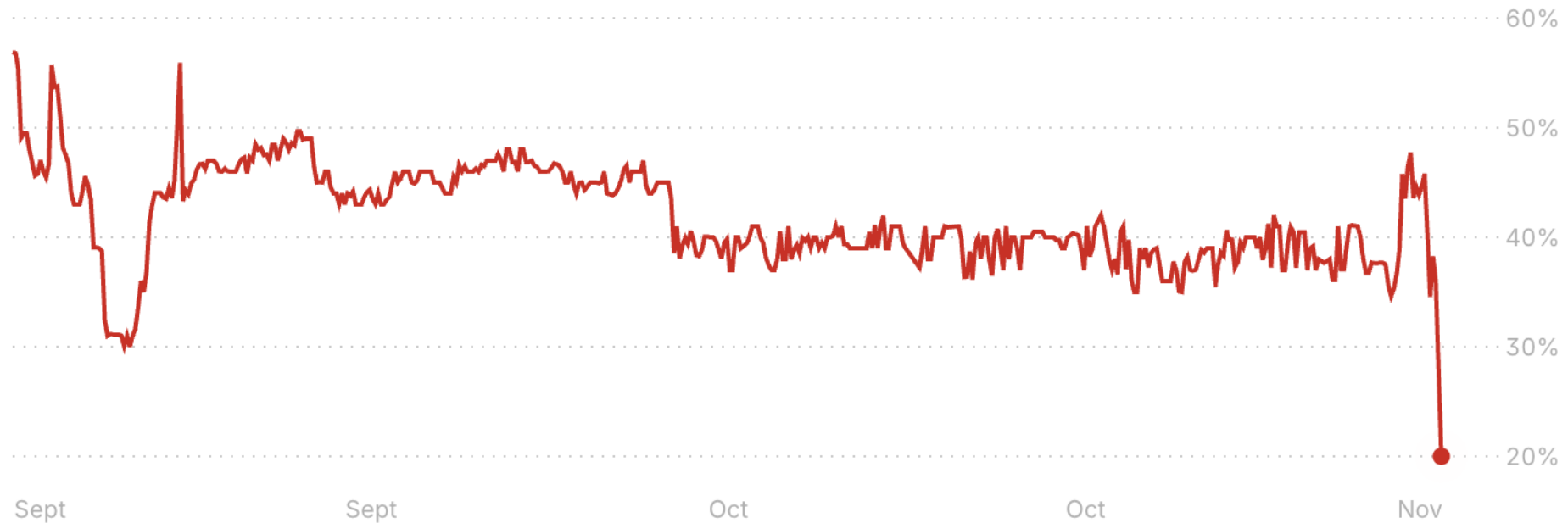


## Will the Supreme Court rule in favor of Trump's tariffs?

53  

20% chance ▼ 36.9 ⓘ

Kalshi



Buy

Sell

Limit

▼

Yes 27¢

No 74¢

Limit Price

- 74¢ +

Shares

510

20.00 matching

-10

+10

Set Expiration

☐

Total

\$377.40

To Win

\$510

Trade



Will the Supreme Court rule in favor of Trump's tariffs?

Buy Yes

Buy

Sell

Dollars

▼

Yes 21¢

No 80¢

Dollars

Earn 3.5% Interest

\$130.00

Odds

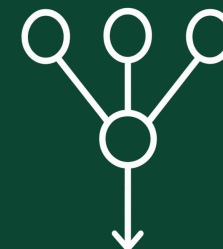
22% chance

Payout if Yes

▼

\$560

Buy



# An Example

**Cost:  $130 + 377.40 = \$507.40$**

**Profit:**

- **If YES: \$560 (10%)**
- **If NO: \$510 (0.5%)**

**How do we even out the results?**

**Quite easy in prediction markets – Each contract is sold for 1\$ (if successful), we should buy the same shares.**



# An Example - Optimized

**Cost:  $\$107.10 + \$377.40 = \$489.6$**

**(510 YES contracts at \$0.21 and 510 NO contracts at \$0.74)**

**Profit:**

- **If YES: \$510 (4%)**
- **If NO: \$510 (4%)**

**Guaranteed 4% profit**

